

Olfactory input to the parahippocampal region of the isolated guinea pig brain reveals weak entorhinal-to-perirhinal interactions

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Abstract

The processing of olfactory inputs by the parahippocampal region has a central role in the organization of memory in mammals. The olfactory input is relayed to the hippocampus via interposed synapses located in the piriform and entorhinal cortices. Whether olfactory afferents directly or indirectly project to other areas of the parahippocampal region beside the entorhinal cortex (EC) is uncertain. We performed an electrophysiological and imaging study of the propagation pattern of the olfactory input carried by the fibres that form the lateral olfactory tract (LOT) into the parahippocampal region of the *in vitro* isolated guinea pig preparation. Laminar analysis was performed on field potential depth profiles recorded with 16-channel silicon probes at different sites of the insular–parahippocampal cortex. The LOT input induced a large amplitude polysynaptic response in the lateral EC. Following appropriate LOT stimulation, a late response generated by the interposed activation of the hippocampus was observed in the medial EC. LOT stimulation did not induce any local response in area 36 of the perirhinal cortex (PRC), while a small amplitude potential with a delay similar to the lateral EC response was inconsistently observed in PRC area 35. No PRC potentials were observed following the responses evoked by LOT stimulation in either the lateral or the medial EC. These findings were substantiated by current source density analysis of PRC laminar profiles. To further verify the absence of EC-to-PRC field interactions after LOT stimulation, high-resolution optical imaging of neuronal activity was performed after perfusion of the isolated brain with the voltage-sensitive dye RH-795. The optical recordings confirmed that olfactory-induced activity in the EC does not induce massive PRC activation. The present findings suggest that the olfactory input into the parahippocampal region is confined to the entorhinal cortex. The results also imply that, as demonstrated for the PRC-to-EC pathway, the propagation of neuronal activity from the EC to the PRC is hindered, possibly by a powerful inhibitory control generated within the EC.

Introduction

The fibres that form the lateral olfactory tract (LOT) convey specific sensorial information generated in the olfactory bulb into the piriform (PC) and entorhinal (EC) cortices. Extensive anatomical and physiological studies demonstrated that such a projection targets the apical dendrites of neurons located in the superficial layers of both the PC and the rostro-lateral part of the EC (Luskin & Price, 1983; Kosel *et al.*, 1981; Biella & de Curtis, 2000; Haberly, 2001; Mouly *et al.*, 1998; Insausti *et al.*, 2002; Habets *et al.*, 1980). The lateral EC (l-EC) represents the gateway through which the olfactory information reaches the dentate gyrus (DG) of the hippocampus, via the perforant path (Wilson & Steward, 1978; Cragg, 1960; Iijima *et al.*, 1996). High intensity or repetitive LOT stimulation at specific rates (1–8 Hz) have been shown to entrain the hippocampal formation to generate a feed-

back projection to the deep layers of the medial and caudal EC (Hjorth-Simonsen, 1972; Biella & de Curtis, 2000). Anatomical evidence also suggests that, in the rat, the ventral bank of the perirhinal cortex (PRC), namely area 35, receives LOT fibres (Burwell *et al.*, 1995; Deacon *et al.*, 1983) and projects either directly (Kosel *et al.*, 1983; Deacon *et al.*, 1983) or indirectly (Naber *et al.*, 2000) to the hippocampus. Previous studies have demonstrated that combined lesions of both the PRC and the EC produce severe impairment of olfactory memory (Otto & Eichenbaum, 1992; Otto & Garruto, 1997; Hanson *et al.*, 2002), suggesting a contribution of the parahippocampal region to olfactory memory. Such a role relies on anatomically well defined cortico-cortical neuronal networks (Swanson & Kohler, 1986; Lopes da Silva *et al.*, 1990; Burwell *et al.*, 1995; Insausti *et al.*, 1997; Burwell & Amaral, 1998b) which are far from being definitively functionally identified. In the present study we characterize with electrophysiological and imaging techniques the pattern of propagation of the olfactory input to the parahippocampal areas in the guinea pig. The experimental approach has been facilitated by the use of the guinea pig isolated brain preparation, which allows for an easy and

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direct visual control of the placement of both the recording electrodes and the imaging apparatus. We confirmed that the l-EC receives an associative olfactory input, whereas we did not observe any direct or EC-mediated synchronous activation of the PRC in response to LOT stimulation.

Materials and methods

Young adult guinea pigs weighing 150–250 g were anaesthetised with pentobarbital (20 mg/kg i.p.) After substituting the blood with a cold (10 °C) oxygenated saline solution (see below) by intracardiac perfusion, the brain was carefully dissected out (de Curtis *et al.*, 1991; de Curtis *et al.*, 1998; Muhlethaler *et al.*, 1993) and transferred to a recording chamber. Brain reperfusion was restored through a polyethylene cannula inserted in the basilar artery with a complex saline solution composed of (in mM) NaCl, 126; KCl, 3; KH₂PO₄, 1.2; MgSO₄, 1.3; CaCl₂, 2.4; NaHCO₃, 26; glucose, 15; HEPES, 2.1; and 3% dextran (M.W. 70 000; Fresenius Kabi, Isola della Scala, Italy), saturated with a 95% O₂/5% CO₂ gas mixture. During the surgical procedure the chamber temperature was maintained at 15 °C and then was slowly raised (0.2 °C/min) to 32 °C before starting the recordings. The experimental protocol has been reviewed and approved by the Committee on Animal Care and Use and by the Ethical Committee of the Istituto Nazionale Neurologico of Milano and the Advanced Industrial Science and Technology of Tsukuba.

Electrophysiological recordings

Field potential laminar profiles were recorded by utilizing multi-channel silicon probes (kindly provided by the Jammille Hetke of the Center of Neural Communication Technology of the Michigan University) which featured 16 independent iridium recording sites, each separated by 50 µm, vertically aligned on a single shaft (input resistance of each recording site between 2 and 3 MΩ). The probe was inserted perpendicular to the cortical lamination at different sites close to the rhinal sulcus. The placement of the recording probe was performed under direct visual control via a stereo microscope (Nikon, SMZ-2T). The position of the probe was further verified by histological identification of either the probe track or the lesion obtained by passing a 20-µA current for 20 s between the deepest contacts of the probe located at 700–750 µm depth in the cortex. Histological controls were performed on 75-µm coronal sections cut from brains fixed with 4% paraformaldehyde and stained with thionin. Bipolar electrodes (custom-made twisted silver wires) were positioned on the LOT for stimulation. Bipolar tungsten electrodes (FHC, Bowdohiman, USA) inserted in the brain according to stereotaxic reference points (Rapisarda & Bacchelli, 1977) were utilized to stimulate the nucleus lateralis of the amygdala.

Averages of 10–15 responses were utilized to build field potential laminar profiles recorded with the 16-channel silicon probes via an AC-coupled extracellular amplifier (Biomedical Engineering, Thornwood, NY). Traces were digitally stored either on tape by means of a DAT recorder (Biologic, France) or on the hard drive of the PC (1000-Hz sampling rate). Current source density (CSD) analysis was performed on 50-µm-per-step profiles with a 200-µm separation grid, as previously described (Biella & de Curtis, 1995). On-line and off-line analysis was performed by utilizing a software developed in LabView graphical language. Special subroutines for the implementation of CSD analysis on laminar field potential profiles were developed in our laboratory.

Optical imaging

In eight experiments, optical recordings of the fluorescent signal generated by the tissue after arterial perfusion with the voltage-

sensitive dye RH-795 (Molecular Probes; 1 mg in 20 mL perfusate for 5–10 min) were performed (see Iijima *et al.*, 1996). A single staining of the brain allowed recording of large optical signals for 2–3 h. A laboratory-designed two-fold epifluorescence microscope, consisting of two lenses ($f = 50$ mm F/1.2 and $f = 100$ mm F/1.85, Nikon, for the first and second objectives, respectively), a dichroic mirror (580 nm) and excitation (535 nm) and absorption (600 nm) filters, was mounted above the parahippocampal region of the isolated guinea pig brain. Emitted fluorescence was collected using a MOS-based monolithic array camera (Ichikawa *et al.*, 1993) and optical responses were presented as 64 × 64-pixel images at a rate of 0.6 ms/frame. A cortical area of 4.2 × 4.2 mm was imaged when the camera was attached to the microscope; each pixel covered 64 µm². For each optical image shown, 8–16 responses were averaged (de Curtis *et al.*, 1999; Takashima *et al.*, 1999).

Results

The experiments were performed on 27 guinea pig isolated brains. The olfactory input to the parahippocampal cortices was mimicked by LOT stimulation. LOT-evoked field potential laminar profiles were recorded with 16-channel silicon probes at different rostro-caudal positions in piriform, entorhinal, perirhinal and insular cortices as described by the scheme in Fig. 1A ($n = 7$). The potentials simultaneously recorded with the 16 leads of the silicon probe between the pial surface and 750 µm depth are illustrated in superimpositions in all figures. According to previous reports (Rodriguez & Haberly, 1989; Ketchum & Haberly, 1993), LOT stimulation evoked a typical mono-disynaptic response in piriform cortex (PC in profiles 6 and 7 of Fig. 1B) and a pure associative polysynaptic response in the l-EC (traces in panels 8 and 9 of Fig. 1B). No depth reversal was observed in the field potential laminar profiles recorded at different rostro-caudal locations lateral to the rhinal sulcus, such as the insular cortex (IC in profiles 1 and 2 in Fig. B) and area 36 of the PRC (panels 3–5 in Fig. 1B). CSD analysis of IC and PRC laminar profiles did not reveal local sinks/sources (not shown). These findings suggest that the responses recorded in the insular cortex and in PRC were not locally generated, but represented volume-conducted far fields possibly generated in the PC/EC (see Discussion). Similarly, no phase reversal was observed when the 16-channel silicon probe was positioned between 750 and 1500 µm depth, to record from PRC deep layers ($n = 5$). After each experiment, the recording location of the 16-channel silicon probes were histologically recognized by identifying the corresponding electrode tracks on 75-µm coronal slices stained with thionine (microphotographs on the left in Fig. 1B).

To better characterize the neural circuits activated by LOT stimulation, subsequent recordings were performed in the same experiment in five adjacent positions located in the PRC and the EC at the same coronal plane ($n = 5$), as schematically illustrated in Fig. 2A and B. As expected, a phase reversal was observed in the superficial layers of the l-EC (profiles 1 and 2). The onset and peak latencies of the locally generated field potentials in the lateral EC were 25.86 ± 0.82 and 36.30 ± 2.00 ms, respectively ($n = 5$). In area 35 of the adjacent PRC (position 3) the presence of a field potential depth reversal, with peak latencies of 31.5 and 29.7 ms, was observed in two out of five experiments (peak latencies of l-EC responses were 33.3 and 32.1 ms in the same experiments). Because of the slow rise of the potentials, onset latencies in area 35 recordings could not be detected with accuracy (see laminar profile 3 in Fig. 2). Neither local potential reversal nor local CSD sinks/sources were observed in the other three experiments, as well as in all area 36 recordings performed in positions 4 and 5. These data suggest that no synchronous synaptic activity is generated

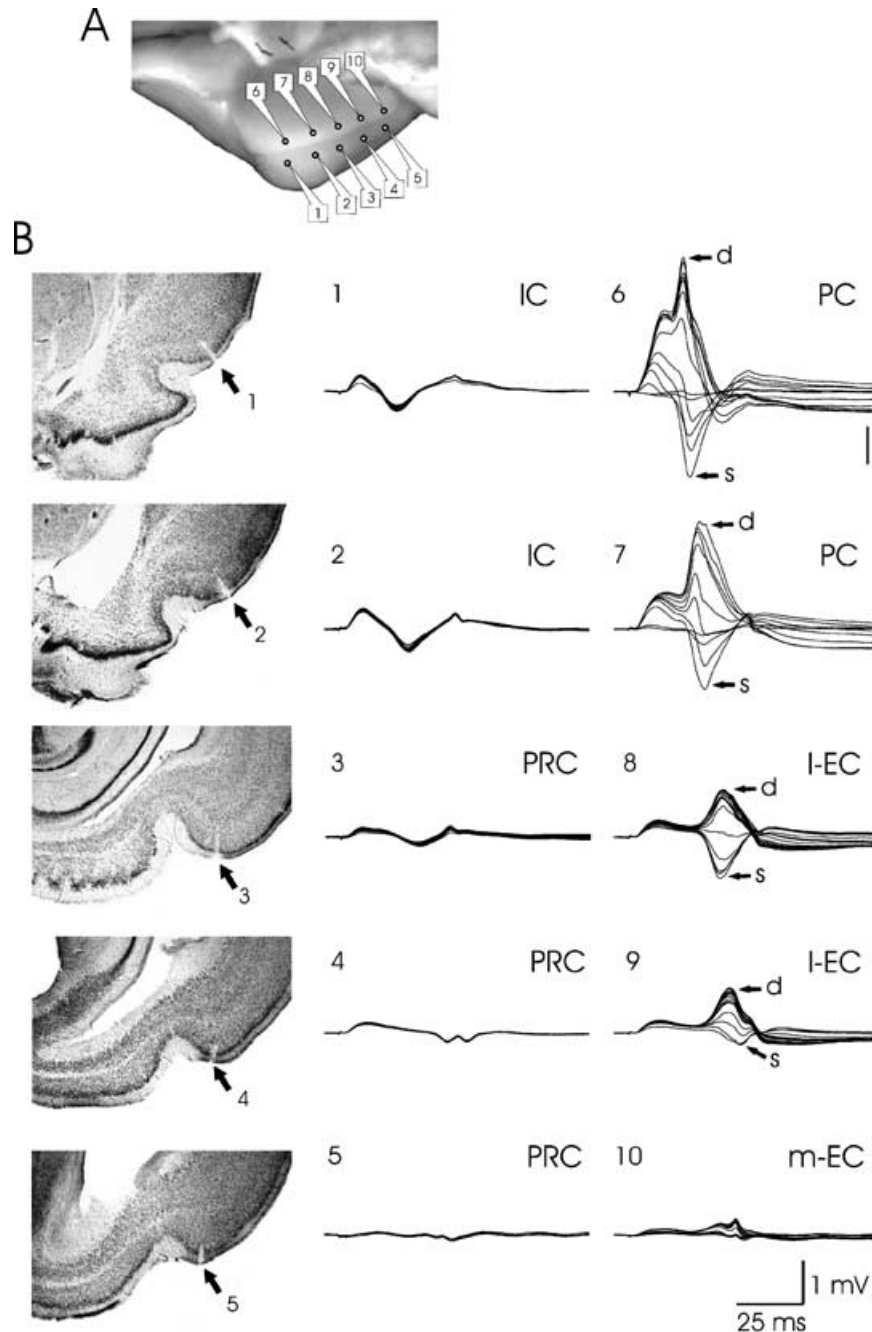


FIG. 1. LOT-evoked field potential profiles in piriform, entorhinal, perirhinal and insular cortices. (A) Ventral view of the right hemisphere of the isolated whole brain. The 16-channel silicon probe was inserted in 10 different positions (black dots), each identified by a corresponding number. The recording sites were arranged in five pairs distributed along the rhinal sulcus at different rostro-caudal levels. (B) Electrode placements in PRC and insular cortex have been verified by the identification of the lesioned tissue (black arrow) on coronal slices stained with thionin. Overlapped LOT-evoked field potential profiles recorded in the 10 positions indicated in A and B. Phase reversal responses were detected only in PC and in EC (6–10). The traces recorded in the superficial and in the deep layers are indicated by the letters s and d, respectively. Only far fields were detected in the corresponding recording sites located in PRC and insular cortex (1–5). IC, insular cortex; I-EC, lateral entorhinal cortex; m-EC, medial entorhinal cortex; PC, piriform cortex; PRC, perirhinal cortex.

within PRC area 36 after LOT stimulation. The absence of a detectable population response in the PRC after the LOT-evoked potential in the I-EC also demonstrates that the activity generated in the I-EC did not propagate massively to the PRC.

The propagation of the olfactory information to the medial EC (m-EC) has been shown to be mediated via the activation of the hippocampal loop (Biella & de Curtis, 2000). To verify whether or not a previous activation of the hippocampal–m-EC circuit is necessary for

transferring olfactory-derived information to the PRC, we simultaneously recorded field potential laminar profiles in the m-EC and in the adjacent PRC following LOT stimulation, as shown in Fig. 3A ($n = 5$). A single shock delivered to the LOT induced in the m-EC the typical associative potential illustrated in panel 1 of Fig. 3B. By appropriately adjusting the intensity and the frequency of LOT stimulation we were able to evoke a delayed m-EC response (profile in 1* of Fig. 3B), which we have previously shown to be generated in deep

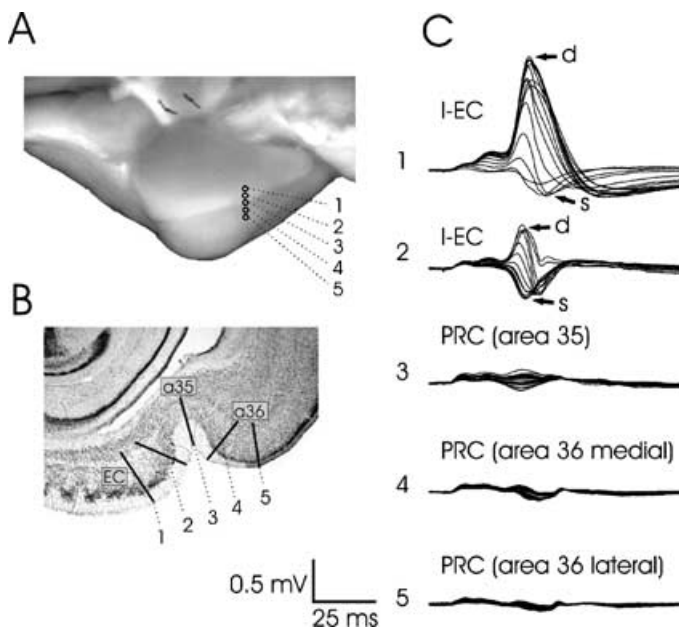


FIG. 2. Extracellular field potentials recorded from five adjacent subregions of PRC and EC located at the same rostro-caudal level. (A) Photomicrograph of the ventral view of the guinea pig brain. The aligned recording sites are represented by the black dots. (B) Portion of the coronal slice stained with thionin. The black lines indicate the five positions of the probe during the different recordings. (C) Overlapped field potential profiles recorded in I-EC (1 and 2) show a clear phase-reversal component. The traces recorded in the superficial and in the deep layers are indicated by the letters s and d, respectively. Field potential profiles recorded in area 35 of the PRC (3) show a small-amplitude potential reversal at ≈ 30 ms delay. In area 36 (4–5) no laminar reversal was observed, suggesting that the responses represent a volume-conducted far field.

m-EC layers by the hippocampal output (see Biella & de Curtis, 2000, for details). No phase reversal was observed in the laminar profile recorded in area 36 of the PRC either in the absence (panel 2 in Fig. 3B) or in the presence (panel 2* in Fig. 3B) of the hippocampus-mediated activation of the m-EC. CSD analysis of area 36 laminar profiles did not reveal local sinks/sources (not shown). To confirm the viability of the PRC in these experiments, the response to stimulation of the nucleus lateralis of the amygdala, which is known to project to PRC (Pelletier & Pare, 2002; Pikkarainen & Pitkanen, 2001; Stefanacci *et al.*, 1996; Pitkanen *et al.*, 2000; Krettek & Price, 1974; Kajiwara *et al.*, 2002), was established as positive control (see inset in panel 2 of Fig. 3B). A clear phase reversal was detectable in the PRC field profile evoked by amygdala stimulation.

The spatial distribution of the olfactory input within the parahippocampal region was further evaluated by analysing the voltage-sensitive dye-derived optical signal evoked by LOT stimulation in the EC and the PRC ($n = 8$). The imaged area in the representative experiment outlined by the dotted line in the upper left panel in Fig. 4 covered a 4.2×4.2 mm cortical surface surrounding the rhinal sulcus. Dots and numbers indicate the pixels from which the optical signals reported in the right panel were measured, in the time domain. According to the electrophysiological recordings (see also Mouly *et al.*, 1998; Biella & de Curtis, 2000) the olfactory-evoked activity spread into the I-EC (traces 1, 2 and 3), from where it diffused to a medial structure (trace 5 in Fig. 4), which correspond either to the location of the temporal pole of the dentate gyrus or to the parasubiculum (see Discussion; Biella & de Curtis, 2002; Rapisarda & Bacchelli, 1977). In five out of eight experiments a late (> 50 ms) activation of the m-EC was detected (traces 5, 6 and 7 in Fig. 4).

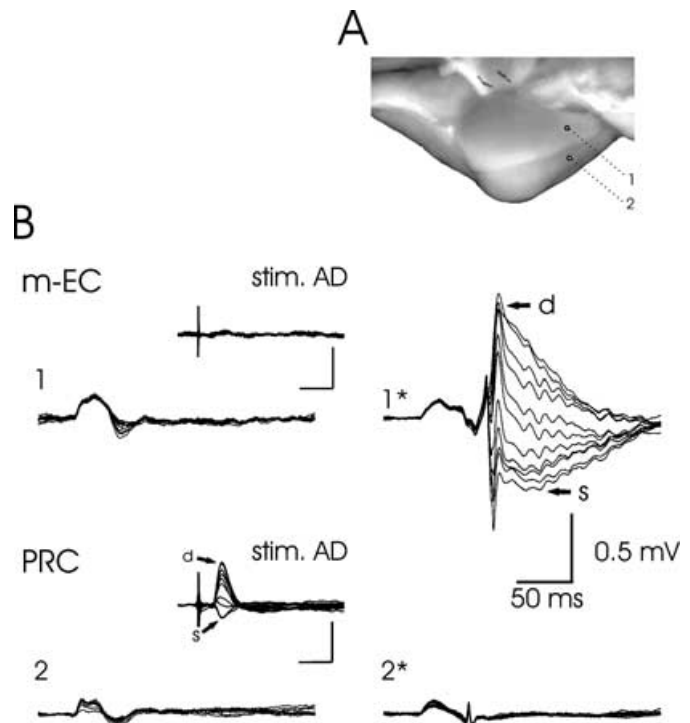


FIG. 3. (A) Schematic drawing of the recording positions in m-EC and area 36 in PRC. Laminar profiles in B1 and B1*, and B2 and B2* were recorded simultaneously in the m-EC and in the PRC, respectively, following LOT stimulation. By changing the intensity of the stimulus, responses with (B1*) or without (B1) a delayed activity in m-EC were evoked. The late m-EC activation was shown to be mediated by the activation of the hippocampus (see Biella & de Curtis, 2000). No local responses were observed in the PRC (B2 and B2*) in the two different conditions of stimulation. As positive control, field potential profiles were recorded in the same positions after stimulation of the nucleus lateralis of the amygdala (AD in the insets in B). The letters s (superficial) and d (deep) on the field potential profiles indicate the position where the traces have been recorded.

No activity was measured at any time delay in the three different positions aligned along the rostro-caudal extent of the PRC (traces 8, 9 and 10 in Fig. 4). The temporo-spatial propagation of neuronal activity at four different time points after LOT stimulation is shown in the lower part of Fig. 4; each panel represents a merging picture between the brain microphotographs and the optical signal detected at the different delays indicated, corresponding to the time points indicated in the upper right panel in Fig. 4.

Discussion

The present study confirms that LOT stimulation activates the I-EC, whereas the m-EC is activated with a longer delay, via the hippocampus (see Biella & de Curtis, 2000). Our experiments demonstrate for the first time (i) that area 36 of the PRC cannot be activated either directly or indirectly by LOT stimulation, and (ii) that the activation of both the lateral and m-EC is not followed by a massive activation of the PRC.

Anatomical data support the view that the olfactory input reaches the most rostral portion of the I-EC. Even though a small contingent of direct fibres which originate in the olfactory bulbs has been shown to terminate in the rostral I-EC (Price, 1973; Wouterlood *et al.*, 1985; Schwerdtfeger *et al.*, 1990; Luskin & Price, 1982), the predominant olfactory afferents to the superficial layers of the I-EC derive from associative cortico-cortical fibres that originate in the piriform cortex

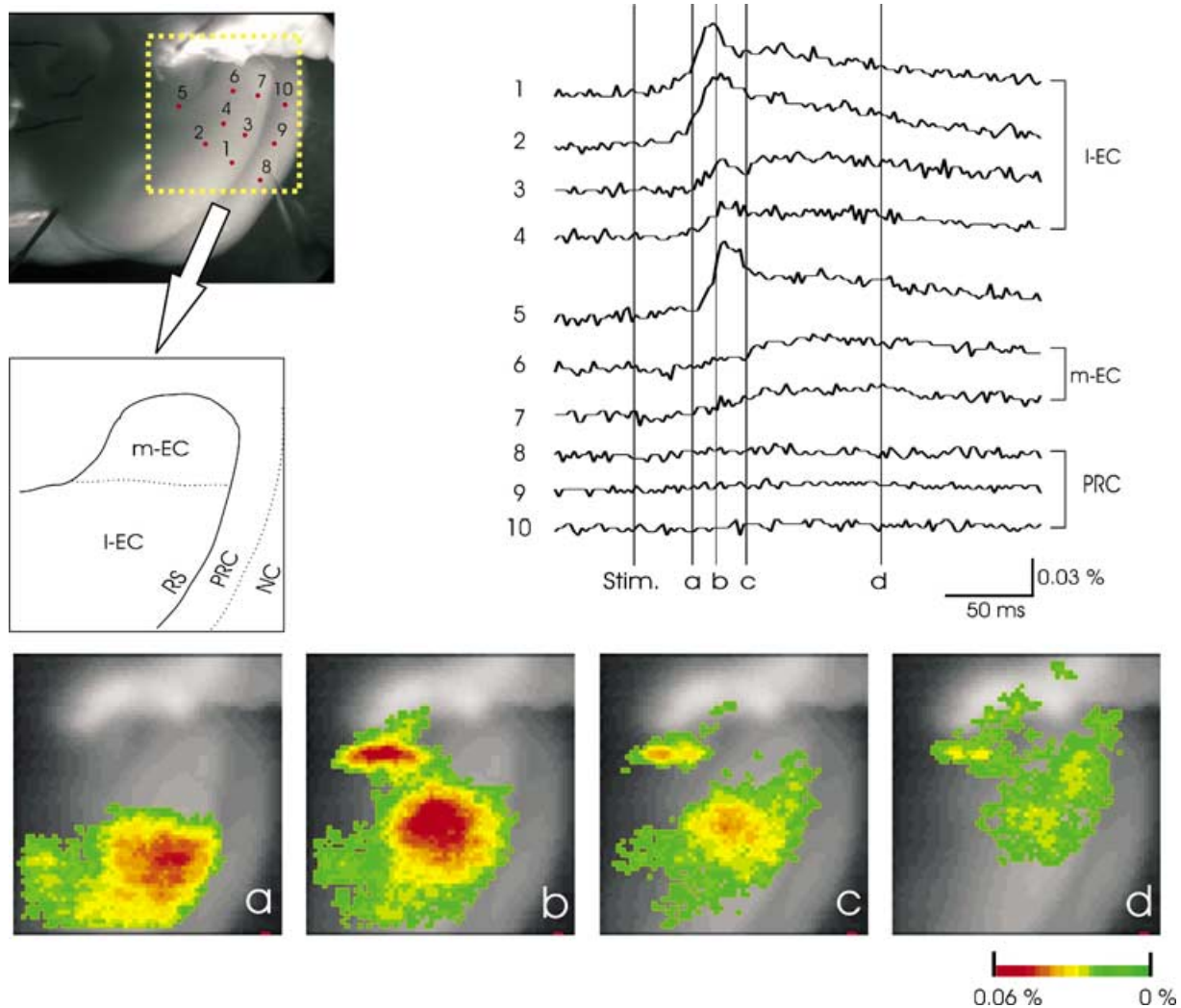


FIG. 4. Optical imaging of the LOT-evoked activity in the EC and the PC. Stimulus intensity was set to induce a late response in the m-EC (see text). A photomicrograph of the ventral view of the isolated whole brain preparation is illustrated in the upper left panel. The optical recording with voltage-sensitive dye was performed in the area indicated by the dotted rectangle. The time-course of the optical activity reported in the right panel was measured as fractional changes in fluorescence intensity of the LOT-evoked optical signals at the 10 pixels illustrated in the left panel (1–10). The electrical stimulation to the LOT occurred at time 0. In the middle left panel a scheme of the borders between different limbic structures under examination is shown. In the bottom panels, LOT-evoked optical changes in the EC are illustrated for the time points marked by the vertical lines of the upper right panel. Fractional fluorescence changes encoded in 'pseudocolour', as indicated in the scale, are superimposed on a bright-field image of the preparation. The sequential activation of the l-EC (panels a–c) and m-EC (panel d) was recorded.

(Price, 1973; Krettek & Price, 1977; Haberly & Price, 1978). The prevalent polysynaptic excitation of the EC by olfactory input was confirmed by electrophysiological experiments aimed at characterizing the activity pattern in l-EC after LOT stimulation *in vivo* (Wilson & Steward, 1978; Boeijinga & Van Groen, 1984; Chapman & Racine, 1997; Mouly *et al.*, 1998) and *in vitro* (Ichikawa *et al.*, 1993; Federico *et al.*, 1994; Biella & de Curtis, 2000). l-EC invasion by the olfactory input has also been observed during natural odour stimulation (Macrides *et al.*, 1982; Chaillan *et al.*, 1999). No clear monosynaptic activation of the l-EC in response to LOT stimulation was observed in our experiments. These findings suggest that the l-EC is a second-order associative cortex with respect to the olfactory input.

Following LOT stimulation we observed neither a direct synchronous population response nor a detectable voltage-sensitive signal in area 36, whereas a field potential which showed a depth reversal was inconsistently observed in area 35, with a delay similar to the response recorded in the l-EC at the same rostro-caudal level. The signal generated by the narrow cortical strip of cortex formed by area 35

located at the fundus of the rhinal fissure could not be resolved with the imaging technique because of the presence of a large-capacity vessel enfolded in the pia mater which attenuated the intensity of the signal detected by the system. The absence of a local field response in area 36 could be due to the fact that the fibre contingent that carries the olfactory input is either not laminarly segregated or very weak. In line with our findings, studies performed with anterograde and retrograde tracers showed a weak direct projection from the olfactory bulb/piriform cortex to the PRC, characterized by a sizeable input into the superficial layers of area 35 located in the depth of the rhinal fissure (Kosel *et al.*, 1981; Deacon *et al.*, 1983; Burwell & Amaral, 1998b), while a very small fibre contingent directed to the superficial layers of the medial part of area 36 was reported by one study only (Burwell & Amaral, 1998b).

Anatomical studies also demonstrated the existence of a projection from the EC to the PRC. Pressure injections of tracers in the l-EC gave prominence to a widespread projection to areas 35 and 36 in the rat (Deacon *et al.*, 1983; Kosel *et al.*, 1981) and in the guinea pig

(Sorensen, 1985). More selective iontophoretic injections of tracers showed that these anatomical connections are very weak, originate from layer III and V neurons in I-EC and terminate in all cortical layers of the area 35 with little, if any, contribution to area 36 (Burwell & Amaral, 1998a). Based on the existence of projection fibres between the I-EC and the PRC, a synaptic activation in the PRC via the I-EC should be expected when the olfactory stimulation massively activates the EC. In spite of this anatomical evidence, we did not observe with either extracellular recordings or optical imaging any delayed associative population activity in PRC after either lateral or m-EC activation. No difference in the propagation of the olfactory signal to the PRC could be detected in experiments in which we stimulated the LOT at different intensity and repetitively at different frequencies ranging from 1 to 100 Hz (not shown). Similar results have been reported utilizing optical imaging of intrinsic signals in the isolated guinea pig EC (Federico *et al.*, 1994). As previously discussed (see de Curtis *et al.*, 1998), technical limitations in resolving low synaptic activity with a widespread laminar distribution may exist when field potential laminar analysis is utilized or when the amount of excitation is below the threshold detection of the optical system. Alternatively, as for the PRC-to-EC connection (Biella *et al.*, 2002), the propagation of activity to the PRC may be dampened by the presence of an inhibitory control of the perirhinal-projecting neurons in the EC. This possibility is corroborated by the demonstration that the propagation of activity from the olfactory cortex to the PRC is facilitated in conditions of hyperexcitability, such as after kindling (McIntyre & Plant, 1993; Kelly & McIntyre, 1996) and during applications of GABA-receptor antagonists in the EC close to the rhinal sulcus (Federico & MacVicar, 1996; M. de Curtis, G. Biella, R. Kajiwarra & I. Takashima, unpublished observations). Interestingly, Young and collaborators (Young *et al.*, 1997) demonstrated single-unit activities in either the I-EC or the dorsal PRC (area 36) *in vivo* during odour sampling and during odour memory tasks, such as recognizing specific olfactory cues, suggesting that single neurons in the PRC are accessible to the olfactory input. To clarify the mechanisms which control the propagation of activity from the EC to the PRC (and vice versa), further combined morphological and electrophysiological studies at cellular level should be performed.

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Abbreviations

CSD, current source density; DG, dentate gyrus; EC, entorhinal cortex; I-EC, lateral EC; LOT, lateral olfactory tract; m-EC, medial EC; PC, piriform cortex; PRC, perirhinal cortex.

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